Numbers and Sets

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1. Proofs

A proof is a sequence of true statements without logical gaps, to establish a conclusion.

We have to start somewhere and have agreed assumptions i.e. ‘axioms’.

Why do we even prove?

1. We want to know for certain a statement is true.
2. We want to gain insight into why they are true.
3. We might be lucky and find a ‘beautiful’ proof.

Example (State) ♢

State

Definition ♣